



CSC432 – INFORMATION SECURITY

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Lecture No 1

COURSE INTRODUCTION AND OVERVIEW

Welcome to CSC432



- ☐ Ever wonder what that line at the bottom of the browser means?
- ☐ What a firewall is?
- ☐ Have you ever been authenticated, and if not, does it hurt?
- ☐ Has your social media account ever been hacked?
- ☐ What is a VPN and do they come in blue?
- ☐ DNS compromised, heard of it?

- ☐ If any of these questions resonate with you or feel familiar, you've come to the right place. **Welcome to CSC432**

Welcome to CSC432



- ☐ Did you encounter anything related to information security
 - ☐ today?
 - ☐ over the last week?
 - ☐ over the semester break?

Welcome to CSC432



•   <https://www.facebook.com/>

- ❑ **https** invokes the Secure Socket Layer (SSL) communication security protocol to securely transmit your data to the server
- ❑ SSL uses cryptography, which is used in information security

Welcome to CSC432



- ☐ Other uses of cryptography
 - ☐ ATM machines
 - ☐ On-line banking
 - ☐ Remote login and file transfer using SSH
 - ☐ Digital rights management
 - ☐ E-commerce
 - ☐ Secret broadcasting

Welcome to CSC432



"The art of war teaches us to rely not on the likelihood of the enemy's not coming, but on our own readiness to receive him; not on the chance of his not attacking, but rather on the fact that we have made our position unassailable."

The Art of War, Sun Tzu

Security



Why is security necessary?

- ☐ Because people aren't always nice
- ☐ Because a lot of money is handled by computers
- ☐ Because a lot of important information is handled by computers
- ☐ Because our society is increasingly dependent on correct operation of computers

Security



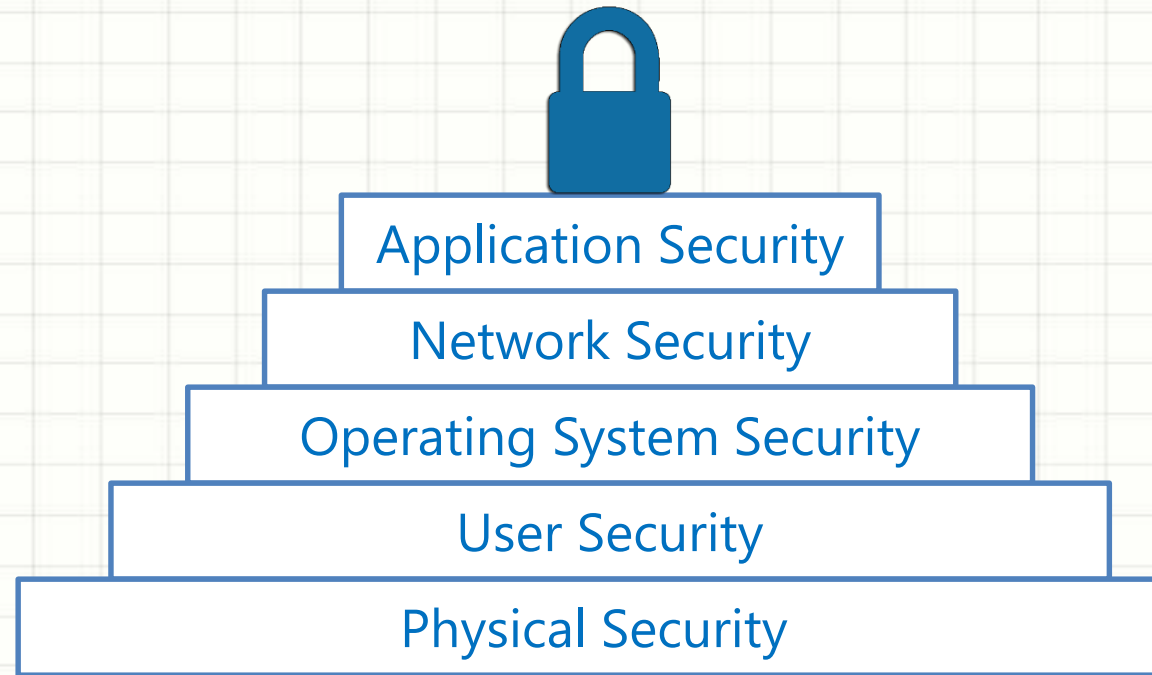
How do security breaches impact organizations?

- ☐ Effectiveness of primary operations are reduced
- ☐ Financial loss
- ☐ Damage to assets
- ☐ Harm to individuals

Security



Layers of security



- ❑ A lock is as strong as the weakest door

Security



- ❑ Security continues to be an issue for organizations
- ❑ Not only do we hear about successful website breaches and network hacks, but we also face new laws and regulations impacting information protection
- ❑ In response to these issues, more and more vendors are appearing with tools that offer some protection and claim that their product was the solution to the security problem
- ❑ From looking at all of this information, it would appear that the big issues in security can be solved with technology
- ❑ Unfortunately, security issues are much more complex than that

Security – First Concepts



- ☐ Lets start with the basics ...
- ☐ So what is security?

Freedom from danger, safety; freedom from fear or anxiety

- ☐ Security in computer terminology

Measures adopted to prevent the unauthorized use, misuse, modification, or denial of use of knowledge, facts, data, or capabilities

OR

Mechanisms to prevent, detect, and recover from network attacks, or for auditing purposes

Security – First Concepts



- ☐ What is a secured computer?
- ☐ A computer is “secured” if it has adequate measures for prevention of, detection of, and recovery from attack(s)

Security – First Concepts



- ❑ At the very bottom, security is a people's issue
- ❑ We can not solve the basic problem with technology, but we can manage the security problem through the dedicated application of well-thought-out security processes and procedures

Security – First Concepts



So what we have learnt

- ☐ You could build the biggest fortress in the world and someone could just come up with a bigger battering ram
- ☐ Security is the name given to the preventative steps you take to guard your information and your capabilities

Security – First Concepts



Security Principles To Follow

- ☐ A 100% secure environment is a myth
- ☐ Expect the unexpected
- ☐ The more you know about someone else, the better you can defend yourself from them
- ☐ The more someone knows about you, the better they can exploit you
- ☐ Do unto yourself before someone does unto you
- ☐ Never assume that all security risks are from the outside, and not the inside
- ☐ Never assume that your users know what they are doing
- ☐ Security is only as good as the weakest link in the chain
- ☐ Security is a process, not an end product

Security – First Concepts



- ☐ Do not forget about the simple aspects of physical security
- ☐ Security through obscurity doesn't work
- ☐ It is a vice to trust all, and equally a vice to trust none

- ☐ Policy first, access second
- ☐ Deny first, permit only when required

Security – First Concepts



Why security is a process, not point products

- ☐ We cannot rely on a single product to provide all of the necessary security for our computer
- ☐ Many different products and types of products are necessary to fully protect an organization's information assets
- ☐ Lets see a few;

Security – First Concepts



- ☐ Anti-virus Software
- ☐ Access Controls
- ☐ Firewalls
- ☐ Smart Cards
- ☐ Biometrics
- ☐ Intrusion Detection
- ☐ Policy Management
- ☐ Vulnerability Scanning
- ☐ Encryption
- ☐ Physical Security Mechanisms

Course Overview and Introduction



About the Course

- ❑ Comprehensive course on information security
 - ❑ Includes both theory and practice
- ❑ Covers a wide range of emerging technologies used in information security paradigm
- ❑ Identify potential security weakness in systems, and possible solutions or countermeasures
- ❑ Comparison of information security standards or algorithms
- ❑ Practice on hacking and anti-hacker techniques
 - ❑ Demonstration on how to use tools to enhance information security

Course Overview and Introduction



What you can get from this course

- ☐ You can get the ability to;
 - ☐ Identify threats
 - ☐ Evaluate security solutions and technologies
 - ☐ Design high-quality solutions
 - ☐ Write clear, complete scheme specifications
- ☐ If nothing else, develop a healthy sense of paranoia!

Course Overview and Introduction



Pre-requisites

- ☐ No (as per your SoS)
- ☐ But I would strongly recommend this course for those who;
 - ☐ are familiar with basic computer networks
 - ☐ are preparing to investigate more details in selected topics and recent developments in computer security
- ☐ This is a theory course! Largely definitions and proofs, although of applied value

Course Overview and Introduction



What You Need

- ☐ Motivation is critical
- ☐ Mathematical Maturity

Your workload

- ☐ 32 lecture meetings, active class participation
- ☐ 4 quizzes/assignments and possibly a project
- ☐ 2 exams (midterm and terminal)
- ☐ A lot of online/book and research paper readings
- ☐ **Expectations**: Attend class, do your assignments, avoid cheating/copying, ask questions and participate!

Security – Basic Tools

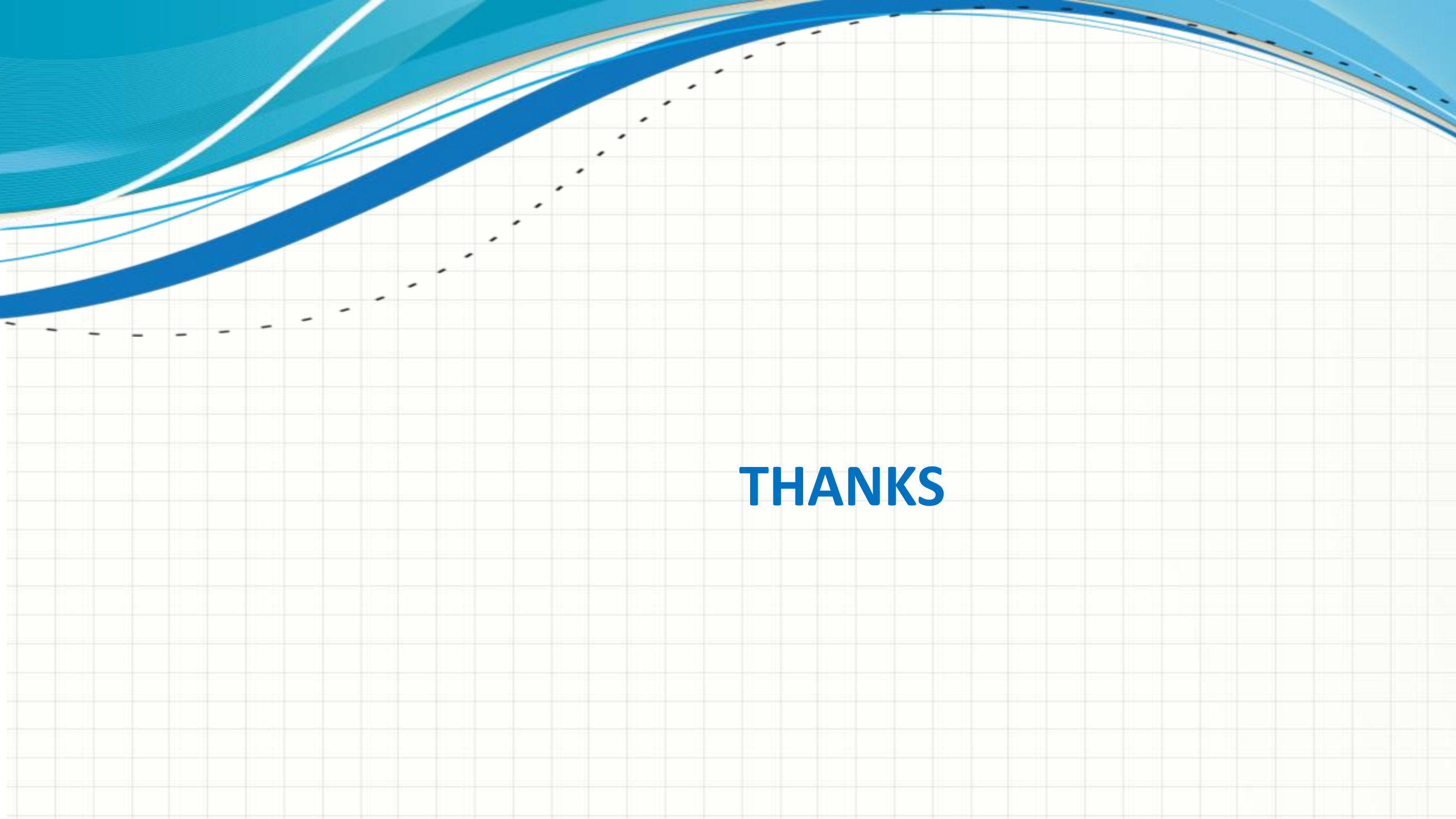


- ☐ Execute the following commands on windows and try all variations:
 - ☐ ipconfig /help
 - ☐ ping /help
 - ☐ arp /help
 - ☐ nslookup
 - ☐ >help
 - ☐ tracert -?
 - ☐ netstat /help
 - ☐ route /help
- ☐ Browse to whois.net and alexa.com, see what these are about
- ☐ Read about "Hosts File" on wikipedia.org

References



- ❑ Center for Education and Research in Information Assurance and Security
<http://www.cerias.purdue.edu/about/history/coast/archive/>
- ❑ Computer and Network Security Reference Index
<http://www.vtcif.telstra.com.au/info/security.html>
- ❑ The Cryptography FAQ
<http://www.faqs.org/faqs/cryptography-faq/>
- ❑ Tom Dunigan's Security page
<http://www.csm.ornl.gov/%7edunigan/security.html>
- ❑ IEEE Technical Committee on Security and Privacy,
<http://www.ieee-security.org/index.html>
- ❑ Computer Security Resource Center
<http://csrc.nist.gov/>



THANKS